Gopalakrishnan Thirunellai Venkitachalam

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EDUCATION

Carnegie Mellon University (CMU) (GPA 4.0/4.0) Pittsburgh, PA
MS in Mechanical Engineering | Specialization: **Artificial Intelligence, Robotics & Motion Planning**
May 2026
Relevant Courses: Generative AI, Advanced NLP, 11785: Deep Learning, Planning & Decision-Making for Robotics

Indian Institute of Technology Madras (IITM) (GPA 8.71/10.0) Chennai, India
B.Tech (**Honors**) in Mechanical Engineering | **Minor** in **Artificial Intelligence and Machine Learning** June 2024
Relevant Courses: Reinforcement Learning, Robotics, Multi-Armed Bandits, Stochastic Processes, Machine Learning

RESEARCH EXPERIENCE

Search Based Planning Lab, Robotics Institute, CMU Pittsburgh, PA
Graduate Research Assistant | Advisor: Prof. Maxim Likhachev Sept. 2024 – Present

- Developing a **real-time heuristic search algorithm** (Constant-Time Planning) for high-dimensional state spaces, optimizing agent trajectories under strict **inference latency constraints (<50ms)**.
- Designing a **generalizable state-abstraction layer** (Policy Transfer) that enables zero-shot adaptation of learned motion primitives across varying environment topologies and constraint manifolds.
- Conducting **ablation studies on search-optimality trade-offs**, analyzing the impact of heuristic inflation on success rates to guarantee bounded computation times in stochastic environments.

PROJECTS

Iterative 2D–3D Inpainting for Single-Image 3D Reconstruction | CMU Sept. 2025 – Dec. 2025

- Architected a **multi-stage generative framework** that integrates **Latent Diffusion priors** (Stable Diffusion) with **foundational depth models** (Depth-Anything-V2) to hallucinate and lift occluded structures from single-view RGB inputs.
- Engineered an **iterative consistency mechanism** that projects 2D generative outputs into 3D space, detecting and repairing high-variance regions to resolve the **"hallucination-depth misalignment"** problem common in zero-shot lifting.
- Outperformed the **state-of-the-art TripoSR** baseline, achieving a **55% reduction in geometric error** (Chamfer: 0.066 → 0.030) and boosting Multi-View Consistency (MVCS) to **0.971**, validating the efficacy of iterative generative refinement.

Latent Test-Time Compute for Generative Retrieval | CMU Sept. 2025 – Dec. 2025

- Investigated **inference-time compute scaling** in Generative Retrieval (RIPOR) by injecting **learnable pause tokens** and encoder masks to induce latent reasoning within hierarchical document identifier generation.
- Conducted a rigorous **ablation study on MS-MARCO**, identifying a critical **beam-search sensitivity bottleneck**: while interleaved computation yielded **+2% Recall@10** in constrained-beam settings, standard decoding strategies failed to generalize to high-recall benchmarks (TREC-DL).
- Analyzed **token-level error distributions**, revealing that imitative models suffer from "early-stage ranking collapse" unlike standard autoregressive models; proposed a **confidence-guided compute allocation** strategy to dynamically target high-entropy decision nodes.

Retrieval-Augmented Generation (RAG) System | CMU Sept. 2025 – Oct. 2025

- Architected a domain-specific RAG pipeline (Firecrawl, PyPDF) with **sentence-aware chunking**, implementing **Reciprocal Rank Fusion (RRF)** to integrate Dense (MiniLM) and Sparse (BM25) retrieval signals.
- Achieved a **statistically significant 3.6× F1 gain (0.12 → 0.43, $p<0.001$)** over the Qwen-2.5-7B baseline on a custom 171-sample benchmark, validating Hybrid RAG superiority for open-domain QA.

Stock Predictor — FinDe++ | CMU Jan. 2025 – June 2025

- Architected a **decoder-only Transformer** using **Time2Vec embeddings** and a novel **Geometric Brownian Motion (GBM)** physics loss, regularizing the model to adhere to stochastic market dynamics while preventing look-ahead bias.
- Integrated **FinBERT sentiment analysis** with technical indicators to capture market regime shifts, demonstrating superior **trend-following capabilities** compared to ARIMA baselines despite discrete sentiment limitations.

SKILLS

Languages: Python (PyTorch, NumPy, Pandas), C++ (STL), Bash
Deep Learning & GenAI: Transformers (Hugging Face), LLM Fine-tuning (LoRA/PEFT), Diffusion Models, RNNs/LSTMs
Retrieval & NLP: RAG (LangChain/Firecrawl), Vector Databases (FAISS/Chroma), BM25, Semantic Search, BERT
Infrastructure & Tools: Linux, Git, Docker, CUDA, Weights & Biases (W&B), Slurm, AWS/GCP